

Apple Inc. (AAPL) Historical Stock Market Analysis Using Python

Week 6 Internship Project – Advanced Python Analysis

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1. Dataset Overview

This project analyzed the historical stock market performance of Apple Inc. (AAPL) using Python. The dataset was obtained from Yahoo Finance and contained approximately five years of daily stock data, including Date, Open, High, Low, Close, Adjusted Close, and Trading Volume. After data cleaning, the dataset contained **1,253 observations** and **7 variables** for analysis.

2. Data Transformation Techniques Applied

The dataset was prepared using Pandas by converting the Date column to datetime format, removing unnecessary columns, converting numeric columns to the correct data types, sorting the data chronologically, and checking for missing values. These transformations ensured the dataset was suitable for time-series analysis.

3. Trends Identified

The time-series analysis showed that Apple's stock price generally followed an upward trend over the analysis period despite short-term fluctuations. The 7-day moving average highlighted recent price movements, while the 30-day moving average showed the long-term trend. Trading volume varied across the period, indicating changes in investor activity, and the daily percentage price change revealed periods of both high and low market volatility.

4. Features Created

Several new features were engineered to improve the analysis, including:

- Daily Price Change
- Percentage Price Change
- 7-Day Moving Average
- 30-Day Moving Average
- Monthly Returns
- 30-Day Volatility
- Year
- Month
- Month Number

These features enabled more detailed trend analysis and performance evaluation.

5. Key Insights

The analysis indicated that Apple maintained steady long-term growth while experiencing normal daily price fluctuations. Periods of increased trading volume often coincided with larger price movements. The moving averages effectively smoothed short-term fluctuations, making overall trends easier to identify. Monthly returns and volatility measures demonstrated that stock performance and market risk varied throughout the year.

6. Recommendations

Investors should combine moving averages, trading volume, and volatility measures when evaluating stock performance. Regular time-series analysis can help identify market trends early and support more informed investment decisions.

7. Conclusion

This project demonstrated the application of Python, Pandas, and Matplotlib for financial data analysis. Through data transformation, feature engineering, time-series analysis, and visualization, meaningful insights into Apple's historical stock performance were generated, strengthening practical skills in Python-based data analytics.

